

XINYI (CINDY) ZHOU

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EDUCATION

University of Southern California

M.S. in Computer Science; GPA: 4.0/4.0

January 2023 – December 2024

Los Angeles, CA

Southern University of Science and Technology

B.E. in Computer Science and Technology; **Outstanding Graduate Award**; GPA: 3.75/4.0

September 2018 – July 2022

Shenzhen, China

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, Java, C#, C++, HTML/CSS

Full-Stack Architecture: Next.js (SSR), Node.js, React, Vue.js; FastAPI, REST APIs, XMPP Integration, Feature Flags (Unleash), Redis

Data/Infra: BigQuery, Snowflake, PostgreSQL, Amplitude, ETL; AWS (ECS, Lambda, S3, RDS/Aurora, CloudFront), Docker, CI/CD

LLMs: PyTorch, Hugging Face Transformers, DeepSpeed, Prompt Engineering, LLM Evaluation, MCP

Developer Operations & Observability: Better Stack, Sentry, Jira, Jenkins, Kibana, Slack Webhooks/Bots, Claude Code, Git Worktrees, AI Agentic Workflows for Automated Error Detection & Debugging

WORK EXPERIENCE

Software Engineer (Backend / AI Product) | Channel AI | Menlo Park, US

Feb 2025 – Present

- Spearheaded an **LLM-powered** personalization feature in **TypeScript**; extended the **PostgreSQL** schema to dynamically inject targeted user context into model prompts, delivering a **+45% revenue uplift** and a 4% increase in daily messaging for mature cohorts within 4 days
- Engineered an automated Trust & Safety pipeline integrating **external ML classifiers**; implemented a **rules-based evaluation engine** that reduced manual review by **78%** by automatically handling definitive cases, routing only failed evaluations to human queues, and logging high-risk content for future policy tuning
- Architected a fault-tolerant, hourly **TypeScript ETL pipeline** routing Amplitude data to **BigQuery**; implemented windowed backfills and automated retries to handle late-arriving events and maintain recommendation engine freshness.
- Created an autonomous Slack bot using **Claude** to surface daily **Sentry/Better Stack** errors and assign owners; enabled conversational debugging that reduced manual log investigation by **~80% per issue** and automatically generated PRs for rapid remediation
- Orchestrated a **time zone-aware**, asynchronous payout pipeline in **TypeScript** to distribute prior-day rewards for **4,000+** creators; implemented **idempotent** retry mechanisms and database state tracking to ensure 100% financial accuracy and system reliability
- Developed scalable backend microservices on **AWS (ECS)** for mobile and web clients, implementing **XMPP** for real-time messaging and **Redis** for session management; optimized Amazon **Aurora PostgreSQL** queries and indexing to significantly reduce latency and enhance data reliability
- Designed and executed **zero-downtime schema migrations** via **Prisma**; utilized **parallel-writing** and **batched backfill scripts** to evolve the **Aurora PostgreSQL** schema without service degradation

Software Engineer Intern | Veeva Systems, Vault Platform Reporting | Pleasanton, United States

May 2024 – August 2024

- Maintained **Spring Boot microservices** and **React** UIs within a high-scale cloud reporting platform; resolved **11 defects** and expanded automated unit and integration test coverage using **JUnit** and **Jest** via **Jenkins** CI/CD pipelines
- Implemented a real-time data freshness indicator by integrating **React** front-end with **downstream microservice APIs**, ensuring accurate visibility into asynchronous data processing states
- Resolved race conditions and key collisions in a **distributed multi-tier caching architecture**, ensuring data consistency across service instances for **1,000+** customers

PROJECTS

Cognitive Biases in LLM-Assisted Software Development | Human-AI Interaction Researcher

January 2024 – October 2025

- Primary researcher for an ICSE 2026 paper on cognitive biases in LLM-assisted software development; led literature review, study design, recruitment, execution, and synthesis (arXiv:2601.08045)
- Ran mixed-methods research with developers (observations: n=14; surveys: n=22), combining behavioral coding and statistical analysis to measure bias in human-LLM workflows
- Built a validated taxonomy of 15 bias categories (systematic analysis of 90 cognitive biases) and translated findings into concrete product/engineering guidelines for designing and evaluating LLM-driven systems with bias-mitigation guardrails
- Measured bias prevalence in human-LLM workflows (48.8% biased actions; 56.4% tied to developer-LLM interactions), using results to define guardrail requirements and evaluation metrics

eBay Search Platform | Full-Stack Developer

September 2023 – December 2023

- Crafted responsive web UIs with **Angular**, **Bootstrap**, **HTML/CSS**, and **JavaScript**, and mobile UIs on Android using **Kotlin**
- Enhanced backend functionality with **NodeJS**, **Express**, **MongoDB**, deployed on **AWS**